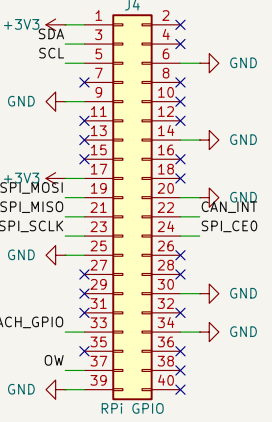
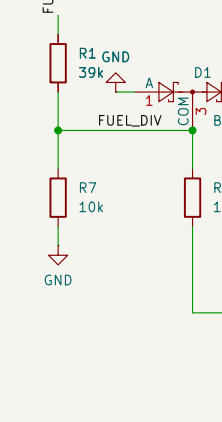


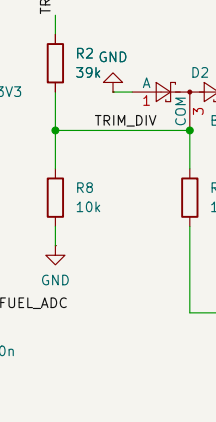
RASPBERRY PI HEADER



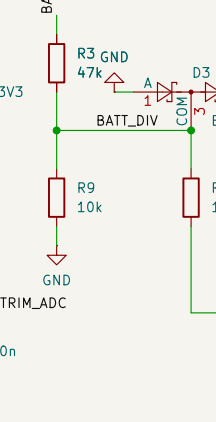
FUEL: J1.1 -> U1 AINO



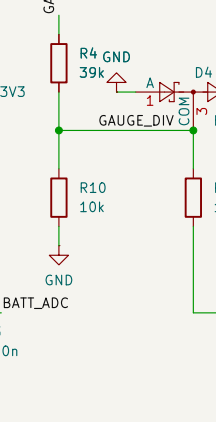
TRIM: J1.2 -> U1 AIN1



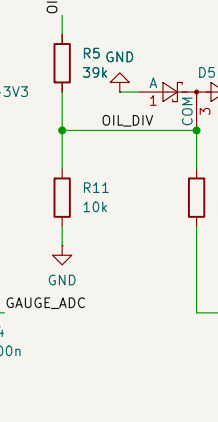
BATT: J1.3 -> U1 AIN2



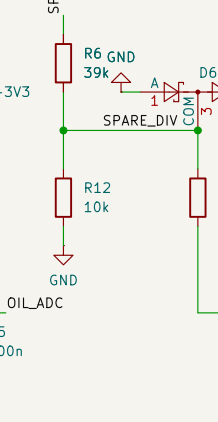
GAUGE: J1.4 -> U1 AIN3



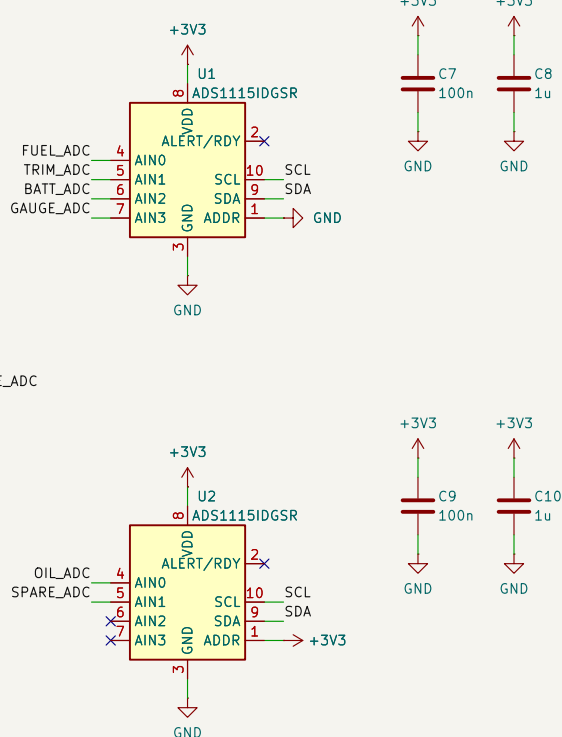
OIL: J1.5 -> U2 AINO



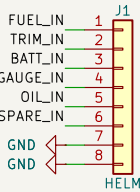
SPARE: J1.6 -> U2 AIN1



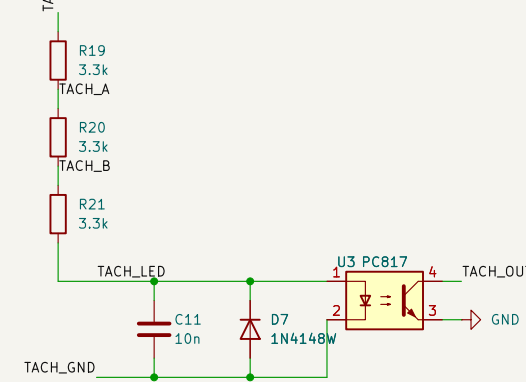
CONVERTERS



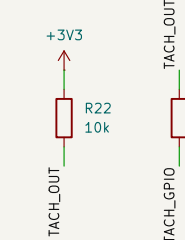
HELM: gauge taps and battery



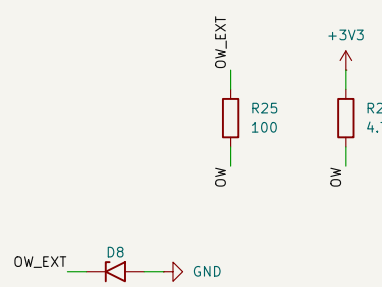
TACH: Delco EST gray wire -> optocoupler



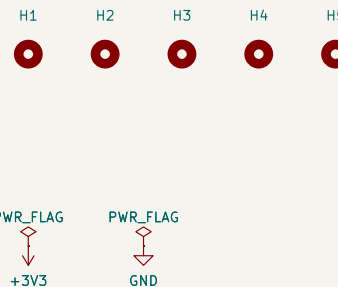
TACH OUTPUT



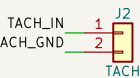
1-WIRE PROBES



MOUNTING / POWER FLAGS



TACH: Delco EST (Ignition side)

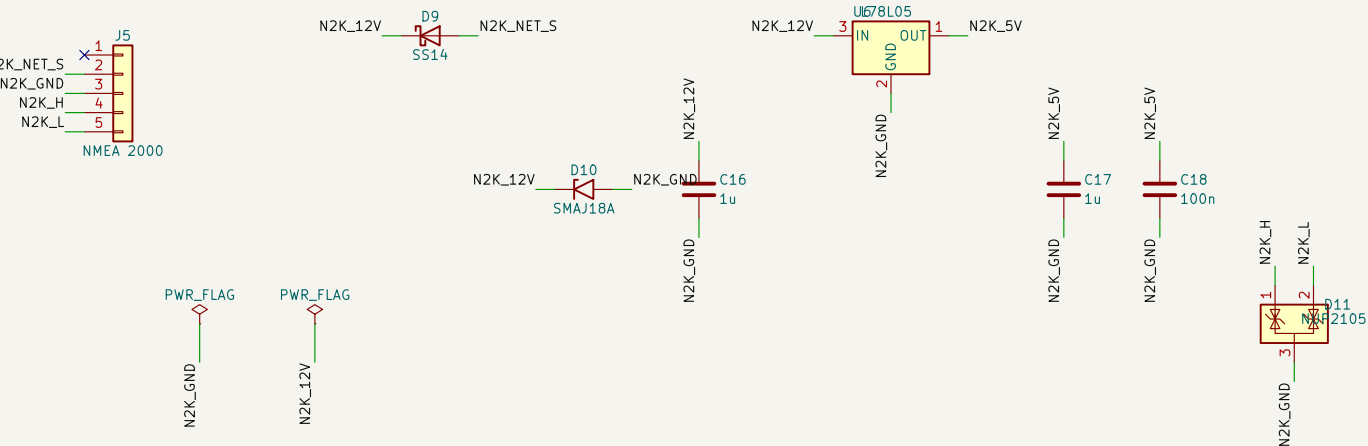


TEMPERATURE PROBES

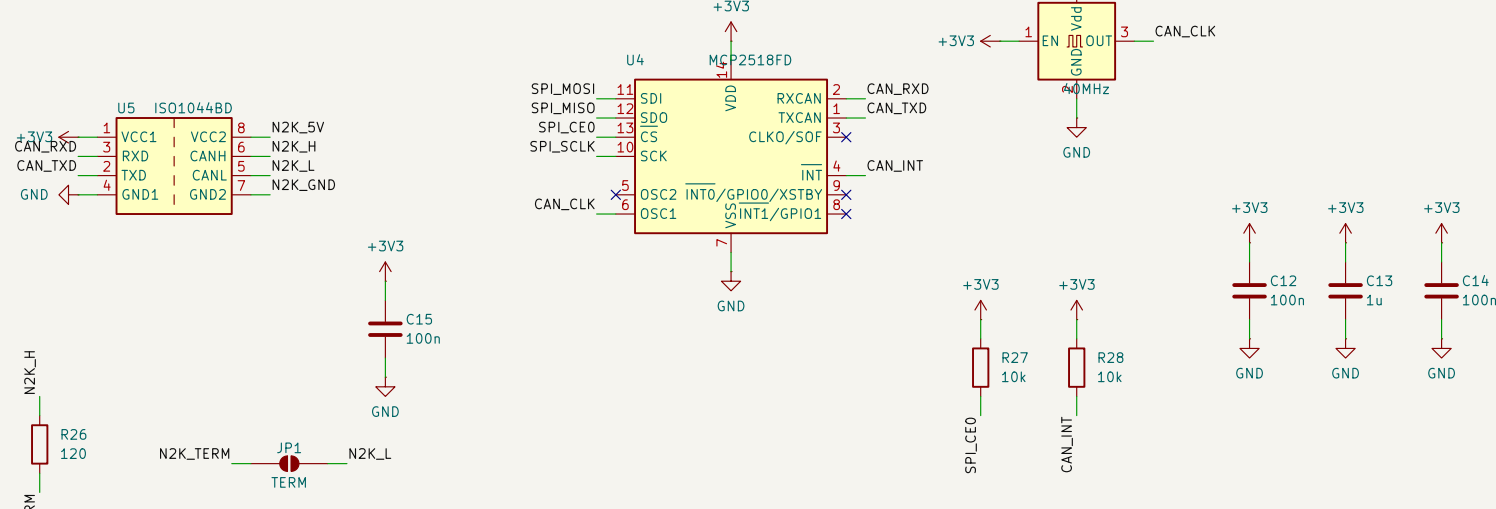


Each gauge tap: the S (or I) terminal -> 10k in heatshink AT THE GAUGE -> J1 -> 39k -> node (10k to GND, BAT54S clamp to GND/3V3) -> 1k -> ADC, with 100n at the ADC pin. V_{adc} = V_{tap} x 10/59 (0-16 V -> 0-2.7 V). Battery: 47k on board, no remote resistor. x 10/57.
Tach, Delco EST ignition (the newer coil: 12 V in, tach out): J2-1 to the gray tach wire, J2-2 to engine ground. The original tach stays on the gray wire.
Takes a 12 V tach signal (~1 mA) or a coil-negative wire (spikes to a few hundred volts); three 3.3k 1206 in series share the spike: 1W4148W and 10n across the LED. One rising edge per ignition cycle on GPIO13; 2 per revolution on the 3.0 4-cylinder (BOAT_TACH_PPR=2).
Keep every TACH_* net (the ignition side) 3 mm from all other copper; see sensor-board.kicad_dru.
Software: BOAT_SENSORS=real BOAT_SENDER_WIRING=tap BOAT_TACH_PULL=none BOAT_TACH_GPIO=13 BOAT_OIL_SENDER=true; dtovrlay=w1-gpio.gpioin=26. Channel map: fuel 0, trim 1, battery 2, gauge supply 3, oil 4 -- see SENSOR_BOARD.md.

NMEA 2000: network side (isolated, powered by NET-S)



NMEA 2000: PI side



NMEA 2000 (Fusion stereo, depth transducer): J5 takes a drop cable -- 1 shield (not connected), 2 NET-S, 3 NET-C, 4 NET-H, 5 NET-L. U5 isolates the network from the Pi: its network side runs from NET-S through D9 (reverse polarity), D10 (surges) and U6 (5 V), and N2K_GND is the network's 0 V, not the Pi's GND. Keep every N2K_* net 2 mm from all other copper (sensor-board.kicad_dru). About 1 LEN. JP1 (open) adds a 120 ohm terminator for a bench test with no backbone -- never bridge it on the boat; the backbone is terminated at its ends. Pi: dtparam=mcp251xfid,spi0-0,oscillator=4000000,interrupt=25 -> can0 at 250 kbit/s; BOAT_CAN=can0.

Generated from design.py by schematic.py; edit those, not this file
Passive taps on the existing analog gauges -- they stay connected and working